0580_w24_ms_42_Q6

Question	Answer	Marks	Partial Marks	Guidance
6(a)	$\begin{pmatrix} 4 \\ -3 \end{pmatrix}$	**2**	**B1** for $\begin{pmatrix} 6 \\ -10 \end{pmatrix}$ or answer $\begin{pmatrix} 4 \\ k \end{pmatrix}$ or $\begin{pmatrix} k \\ -3 \end{pmatrix}$	Penalty 1 mark if fraction line in vector
6(b)(i)	(–4, –1)	**1**		
6(b)(ii)	7.21 or 7.211…	**2**	**M1** for $(-6)^2 + 4^2$	Condone -6^2 if recovery seen Accept $\sqrt{52}$ or $2\sqrt{13}$ as final answer
6(c)(i)(a)	$2\mathbf{c} - \mathbf{a}$	**1**		Accept e.g. $-1\mathbf{a} + 2\mathbf{c}$
6(c)(i)(b)	$\mathbf{c} - \mathbf{a}$	**1**		Accept e.g. $-1\mathbf{a} + 1\mathbf{c}$
6(c)(i)(c)	$\frac{1}{5}(\mathbf{a} + 8\mathbf{c})$ final answer	**2**	**M1** for $[\overrightarrow{AP} =]\ \frac{4}{5} \times$ *their*$(2\mathbf{c} - \mathbf{a})$ or $[\overrightarrow{BP} =]\ \frac{1}{5} \times -$ *their* $(2\mathbf{c} - \mathbf{a})$ or for a correct vector route using the lines on the diagram	For 2 marks, accept e.g. $\frac{1}{5}\mathbf{a} + \frac{8}{5}\mathbf{c}$, $0.2\mathbf{a} + 1.6\mathbf{c}$ Condone mixed notation for vector route e.g. $2\mathbf{c} + BP$
6(c)(i)(d)	$\frac{4}{5}(-\mathbf{a} + \mathbf{c})$ final answer	**2**	**M1** for $[\overrightarrow{QP} =] -\frac{4}{5}\mathbf{c} + \frac{4}{5} \times$ *their*$(2\mathbf{c} - \mathbf{a})$ or for a correct vector route	For 2 marks, accept e.g. $-\frac{4}{5}\mathbf{a} + \frac{4}{5}\mathbf{c}$, $-0.8\mathbf{a} + 0.8\mathbf{c}$ Condone mixed notation for vector route - may use *their* (c)(i)(c) in this route e.g. $QA + -\mathbf{a} +$ *their* OP
6(c)(ii)	[QP is] parallel [to CB] $QP = \frac{4}{5}CB$ oe	**2**	**Dep both statements** consistent with *their* (c)(i)(b) and *their* (c)(i)(d) and both vectors in terms of **a** and **c** **B1** for each dep on statement consistent with *their* (c)(i)(b) and *their* (c)(i)(d) and both vectors in terms of **a** and **c**	Accept e.g. $QP \parallel CB$, $QP : CB = 4 : 5$, $\overrightarrow{QP} = \frac{4}{5}\overrightarrow{CB}$ Do not accept contradictory statements e.g. they are parallel and perpendicular Do not accept statements if they are not consistent with *their* (c)(i)(b) and (c)(i)(d)